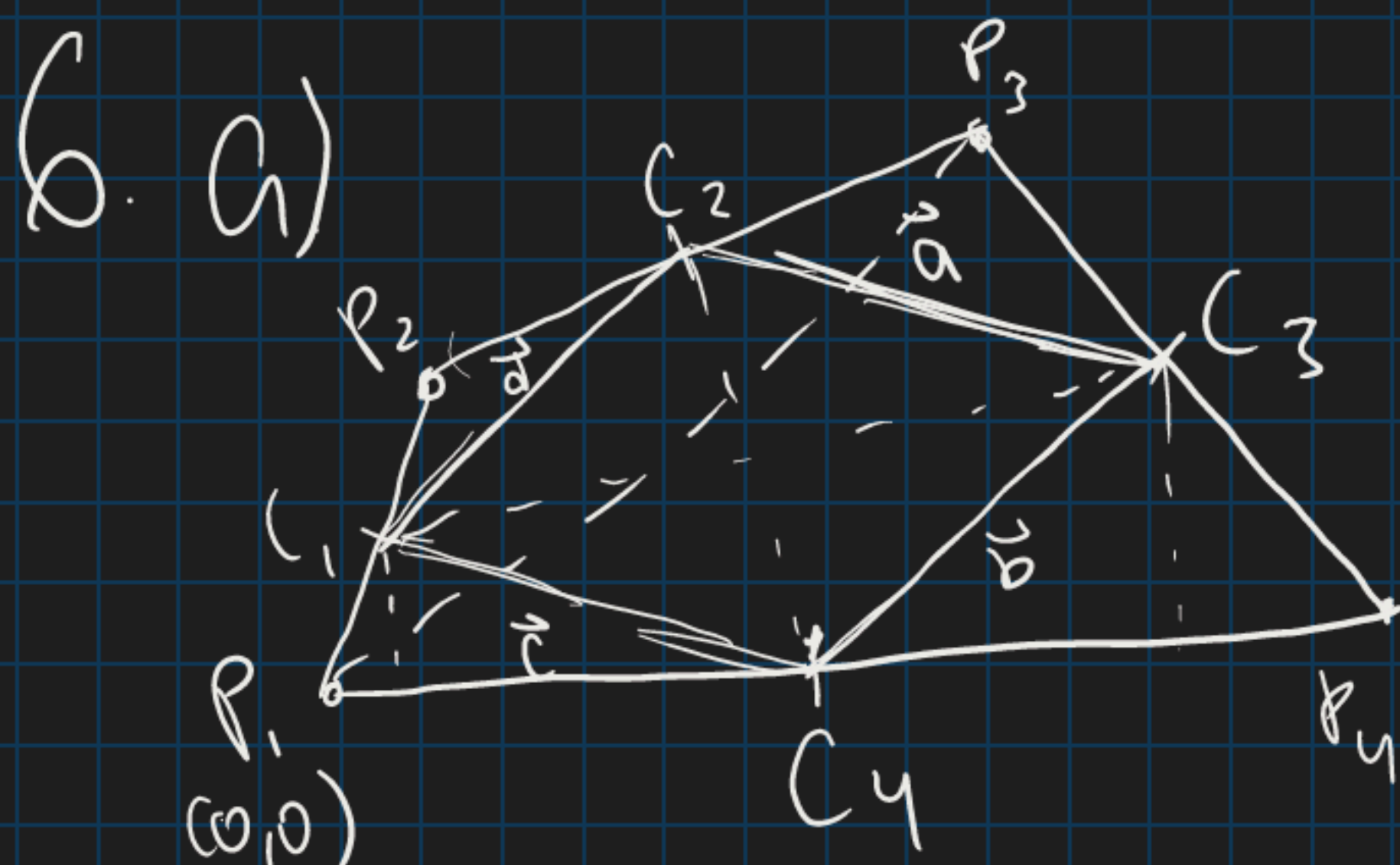
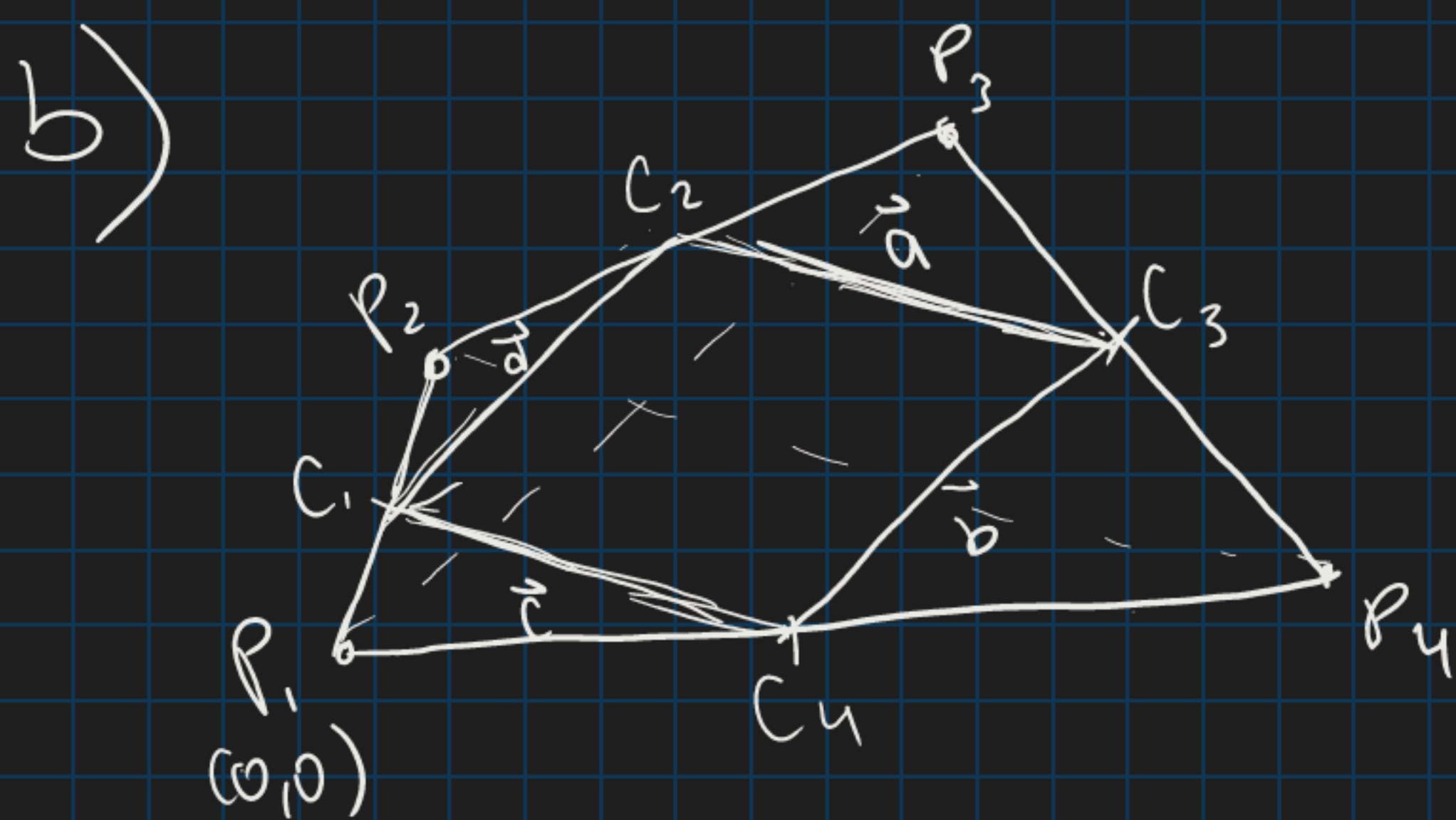




$\Delta P_2 C_2 C_1$ is similar to $\Delta P_2 P_3 P_1$ as

B/c $\Delta P_2 C_2 C_1$ and $\Delta P_4 C_3 C_4$ are both similar to triangles along line $P_1 P_3$, $\vec{a}$ and $\vec{b}$ must be parallel. Therefore the midpoints make a parallelogram



Parallelogram perimeter = sum of ^{quadrilateral's} diagonals

C_1 is the midpoint of $P_2 P_1$, meaning $P_1 P_2$ is double the length of $P_2 C_1$. $\therefore$ similar triangles $P_2 C_1 P_3$ and $P_2 P_1 P_3$ differ by a factor of 2.
 $2 * d = P_1 P_3$. The same can be said for similar triangles $P_3 C_2 P_4$ and $P_3 P_2 P_4$, so $a * 2 = P_2 P_4$.
 $\therefore$ the parallelogram's perimeter = sum of the quadrilateral's diagonals.